

LMS Hub — Update Guide

Update 1.3.0 · package `lmshub-update-2026-10-05.zip`

This update adds four things you asked for and fixes several problems we found while building them. It is a drop-in update: you copy the files over your current installation, install dependencies, run the database migrations and rebuild. Nothing in your existing data is deleted or renamed.

1. What's new

A. Footer pages you can manage (About, Help Center, Privacy, Terms, Contact)

- New screen **Admin Panel → Settings → Pages**: create, edit, publish/unpublish and delete static pages, with a rich-text editor (and an HTML mode).
- Every page has a title, an address (`/pages/<slug>`), a type, a status (Draft / Published / Archived), and two footer options: *Show in footer* and *Footer order*.
- The footer's **Support** column is now built from the pages marked *Show in footer*. The links that used to point to `#` now open real pages.
- Page content is cleaned on the server before it is saved (scripts, event handlers, frames and `javascript:` links are removed) and again in the browser when it is shown.
- The update creates starter pages **About Us, Help Center, Privacy Policy, Terms & Conditions, Contact** and publishes them, so no footer link leads to a "not found" page. Starter text is generic: please review it. Pages you had already written are not touched.

B. Hero image upload

- **Admin Panel → Website → Hero Section → Upload image** now uploads the file directly (the same way the Media Library does), instead of sending it as text inside a JSON request.
- Accepted: JPG, PNG, WebP up to **5 MB** (was 2 MB). The file type is checked from the file's content, not only from its name.
- If your server still refuses the upload, the screen now tells you it was the web server (HTTP 413) and what to change — see section 3.
- Settings → Brand identity now links to Website and Pages, because many admins looked for the hero image there.

C. Super Admin can change their own email

- **Admin Panel → Profile → Change email** (visible to the Super Admin only). You confirm with your current password; the new address must not belong to another account.
- Other users keep the current rule: an administrator changes their email in **Users → Edit**.
- Server command for when you cannot sign in any more: `npm run admin:set-email -- <current-email> <new-email>` (section 5.C).

D. Final exam, passing score, retakes, certificate rules, course restart

- **Final exam per course.** In the course editor, the new tab **Completion & certificate** lets you choose which quiz is the final exam, or none.
- **Passing score (%)** per quiz. $\text{Score} = \text{points earned} \div \text{total points} \times 100$. Each question has a point value (its weight); with 1 point per question this is simply correct answers ÷ number of questions.
- **Certificate locked until the exam is passed.** The certificate unlocks when the student's best graded attempt reaches the passing score. **Courses without a final exam behave exactly as before** (lesson progress only).
- **Attempts:** maximum attempts per quiz (**0 = unlimited**) and an optional **wait between attempts**

(minutes). The student's quiz list shows attempts left and when they may retry.

- **Restart course:** per-course option that lets students reset their lesson progress and study again. Used exam attempts are not given back, and an issued certificate stays valid.
- The existing settings **Minimum progress to pass** and **Minimum passing score** (Settings, group **Certificates**) are now actually applied (they were shown but ignored before).

Fixes included in this update

- Final exams could not be taken after the last lesson was completed (the enrolment became "completed" and quizzes were refused). Fixed.
- Creating a question bank, a quiz or an assignment from the Admin Panel failed (the screens sent field names the API does not accept; "Course ID" had to be typed by hand). Fixed: you now pick the course from a list.
- There was no screen to add questions to a question bank. Added: **Assessments → Question Banks → Questions**.
- The quiz builder did not save the selected questions. Fixed.
- The last answer chosen just before pressing **Submit** could be lost. Fixed.
- Static pages could be edited by any account with the *content* permission, which instructors have (for the Media Library). Pages now require the **Settings** permission, the same as the Website screen.

2. Compatibility

- Requires the same environment as 1.2.0: Node.js 20+, PostgreSQL 14+.
- **Backend dependency added:** `sanitize-html` (run `npm ci`, step 4.4).
- **Database:** two new migrations, both additive with a rollback:
 - `0230_content_pages_footer` — adds `content_pages.tampil_di_footer` and `content_pages.urutan_footer`, creates/publishes the five starter pages (only untouched placeholders).
 - `0240_course_completion_rules` — adds `courses.ujian_akhir_quiz_id`, `courses.izinkan_restart`, `quizzes.jeda_ulang_menit`, and relaxes the check on `quizzes.attempt_maksimal` from " ≥ 1 " to " ≥ 0 " (0 = unlimited).
- Existing quizzes, attempts and issued certificates are unchanged. No course gets a final exam until you choose one.
- The mobile app keeps working; the API only gained fields and endpoints.

3. Before you start

1. **Back up the database:**
`pg_dump -Fc -d "$DATABASE_URL" -f lmshub-before-1.3.0.dump`
2. **Back up the files:** both folders `lmshub-be/` and `lmshub-fe/`, including `lmshub-be/.env`, `lmshub-be/uploads/` and, if you edited it on the server, the deployed `config.js` of the frontend.
3. **Web server upload limit.** For hero images up to 5 MB, the `/api/` location in Nginx needs `client_max_body_size` of at least `6m` (you probably already have a larger value for the Media Library). Example:

```
location /api/ {
    client_max_body_size 2100m;    # Media Library; must be  $\geq 6m$  for the hero image
    proxy_pass http://127.0.0.1:4000;
}
```

4. Update steps

1. **Stop the backend** (for example `pm2 stop lmshub-be` or `sudo systemctl stop lmshub-be`).
2. **Extract** `lmshub-update-2026-10-05.zip`.
3. **Copy over your installation:** copy the contents of `lmshub-update-2026-10-05/lmshub-be/` into your `lmshub-be/` folder and `lmshub-update-2026-10-05/lmshub-fe/` into your `lmshub-fe/` folder, replacing existing files. Do not delete anything; the package only contains changed and new files. (Optional: verify with `sha256sum -c SHA256SUMS.txt` inside the extracted folder.)

4. Backend:

```
cd lmshub-be
npm ci           # installs the new dependency sanitize-html
npm run migrate:up # applies 0230 and 0240
npm run build
```

5. Frontend:

```
cd lmshub-fe
npm ci
npm run build      # then deploy dist/ as you normally do
```

The build writes a fresh `dist/config.js`. If you had changed `apiUrl` in the deployed `config.js`, put your value back after deploying.

6. **Start the backend again** (`pm2 start lmshub-be` / `sudo systemctl start lmshub-be`) and reload Nginx if you changed it.
7. **Clear caches:** reload the site with Ctrl+F5 (or purge your CDN).
8. **Check:** open `/pages/about-us` on your site, sign in as Super Admin and open **Settings → Pages**.

5. Using the new features

A. Pages and footer links

1. Sign in as Super Admin (or a role with the Settings permission).
2. Sidebar **Settings → Pages**. The five starter pages are listed.
3. Press **Edit** on a page. Change the **Title**, the **Address (slug)** if you want a different URL, and write the **Content** with the editor (use **HTML** to paste markup).
4. Set **Status** to **Published** so visitors can open it. Tick **Show in footer** and set **Footer order** (1 = first) to place it in the footer's Support column.
5. Press **Save**. The footer updates immediately. Use **View** to open the public page.
6. **New page** creates another page; **Unpublish** hides a page without deleting it; **Delete** removes it.

If you built your own footer columns in **Website → Footer**, those columns are shown instead of the default ones. Add links such as `/pages/about-us` there.

B. Hero image

1. Sidebar **Settings → Website**, section **Hero Section**.
2. Under **Hero image**, press **Upload image** and choose a JPG, PNG or WebP file (max 5 MB). The upload

progress is shown and the preview updates.

3. **Remove** clears the image (the gradient block is shown instead).
4. If you see "The server refused the file because it is too large (HTTP 413)", raise `client_max_body_size` as shown in section 3.

C. Super Admin email

From the Admin Panel

1. Top-right account menu → **Profile** (page "My Profile").
2. Under **Email**, press **Change email**.
3. Enter the **New email address** and your **Current password**, then **Save new email**.
4. Use the new address the next time you sign in. Your password does not change.

From the server (when you cannot sign in)

```
cd lms-hub-be
npm run admin:set-email -- old-address@example.com new-address@example.com
```

Only a Super Admin account can be changed this way; the command refuses an address that another account already uses. The change is recorded in the audit log.

D. Final exam and certificate rules

1. Write the questions

1. Sidebar **Assessments**, tab **Question Banks** → **+ New Question Bank**. Enter a name, choose the **Course**, **Save**.
2. Press **Questions** on the bank → **Add question**.
3. Choose the **Question type**: single choice, multiple choice (several correct), true/false, or short answer. Write the question, the options, and mark the correct one(s) with ✓. **Points** is the weight of the question.
4. **Save**. Repeat for every question.

Use automatically graded types in a final exam. Essay and file-upload answers are not graded automatically, so an attempt that contains them never receives a final score.

2. Create the exam

1. **Assessments**, tab **Quizzes & Exams** → **+ Create Quiz**.
2. Fill in the title and choose the **Course**.
3. **Passing Score (%)** — for example 75. Leave it empty to use the default from Settings (group Certificates).
4. **Maximum Attempts** — for example 3; **0 = unlimited**.
5. **Wait between attempts (minutes)** — for example 60; 0 = the student may retry immediately.
6. **Time Limit (minutes)** — 0 = no limit.
7. Tick **Final exam of this course**.
8. Choose the **Question bank** and tick the questions to include.
9. **Save Quiz**.

3. Course completion rules

1. **Courses** → open the course → tab **Completion & certificate**.

2. **Final exam** — choose the quiz (or "No final exam" to give the certificate after the lessons, as before).
3. **Allow students to restart this course** — tick to show a **Restart course** button to students.
4. **Save**. This does not send a published course back to review.

4. Defaults for all courses — Sidebar **Settings** → **Settings**, group **Certificates**:

- *Minimum progress to pass* — lesson progress required for a certificate (default 100%).
- *Minimum passing score* — used for quizzes that leave **Passing score** empty (default 70%).

What the student sees

- After the last lesson, the course player shows **Take the final exam**.
- **Assessments → Quizzes & Exams** shows the final exam, the attempts left and, while waiting, when a retake becomes possible.
- **My Certificates** explains why a certificate is still locked ("Your best score is 50% — 75% is needed").
- When the best graded attempt reaches the passing score, the certificate can be issued and downloaded.

Rules in detail

Rule	Behaviour
Score	Points earned ÷ total points of the quiz × 100, two decimals
Passing	Best graded attempt ≥ passing score of the quiz (or the default)
Max attempts	Counts every started attempt; 0 = unlimited
Wait between attempts	Counted from the last submitted attempt
Restart course	Lesson progress → 0%; notes, exam attempts and issued certificates are kept
Exam deactivated or deleted	No longer required for the certificate
Exam added later	Students who have not received their certificate yet must pass it; issued certificates stay valid

6. Rollback

Recommended: restore the backups from section 3:

```
pg_restore --clean --if-exists -d "$DATABASE_URL" lmshub-before-1.3.0.dump
```

then put back the backed-up `lmshub-be/` and `lmshub-fe/` folders, run `npm ci` and `npm run build` in both, and restart.

Without restoring the database (keeps data created after the update):

```
cd lmshub-be
npm run migrate:down # reverts 0240_course_completion_rules
npm run migrate:down # reverts 0230_content_pages_footer
```

Then restore the previous files from your backup, run `npm ci` and `npm run build` in both folders, and restart. What the rollback does:

- Final-exam choices, the restart option and wait times are removed. Quizzes set to "unlimited" (0) get 32767 attempts, which the old version can store.

- Starter pages that were created by the update and never edited are deleted; untouched placeholders go back to Draft. Pages you edited stay.
- Questions, quizzes, attempts and certificates are kept.

7. Known limitations

- Essay and file-upload questions are not graded automatically, so do not use them in a final exam.
- The matching question type is supported by the API and by grading, but it has no editor screen yet.
- *Minimum attendance* (Settings, group Certificates) is still not applied.
- An administrator can reset a student's progress only through the API (`POST /api/v1/enrollments/{id}/restart`); there is no button for it yet.

8. API changes (for integrations and the mobile app)

New:

- `GET /api/v1/public/pages` — published pages (no content), for footers.
- `DELETE /api/v1/pages/:id` — delete a page.
- `POST /api/v1/site-content/asset/hero/upload` — raw image body (`Content-Type: image/jpeg|png|webp`), max 5 MB.
- `PATCH /api/v1/users/me/email` — `{ email, password_saat_ini }`, Super Admin only.
- `PUT /api/v1/courses/:id/completion-rules` — `{ ujian_akhir_quiz_id, izinkan_restart }`.
- `POST /api/v1/enrollments/:id/restart`.

Changed (additive):

- `POST/PUT /api/v1/pages` accept `konten_html`, `slug` (on update), `tampil_di_footer`, `urutan_footer`. Permission is now *Settings* instead of *Content*.
- Quizzes accept `attempt_maksimal: 0` and `jeda_ulang_menit`.
- `GET /api/v1/quizzes/:id` also returns `questions` and `is_ujian_akhir`; the student quiz list returns `is_ujian_akhir`, `nilai_lulus`, `skor_terbaik_persen`, `siswa_percobaan`, `boleh_mulai`, `boleh_lagi_at`.
- `GET /api/v1/question-banks/:id/questions` includes each question's options.
- `GET /api/v1/courses/:id/learn` returns `izinkan_restart` and `ujian_akhir`.
- Certificate claim may answer `409 certificate.final_exam_not_passed` with the exam result in `error.details`; starting an attempt may answer `409 quiz.retake_cooldown` with `boleh_lagi_at`.

9. Files in this package

Backend — 42 files:

```
lmshub-be/migrations/0230_content_pages_footer.ts    (new)
lmshub-be/migrations/0240_course_completion_rules.ts  (new)
lmshub-be/package.json
lmshub-be/package-lock.json
lmshub-be/README.md
lmshub-be/scripts/feature-test-1.3.cjs    (new)
lmshub-be/src/core/html/sanitize.ts    (new)
lmshub-be/src/core/upload/image.ts    (new)
lmshub-be/src/modules/assessments/assessments.controller.ts
lmshub-be/src/modules/assessments/assessments.repository.ts
lmshub-be/src/modules/assessments/assessments.service.ts
lmshub-be/src/modules/assessments/assessments.validation.ts
lmshub-be/src/modules/certificates/certificates.repository.ts
lmshub-be/src/modules/certificates/certificates.service.ts
lmshub-be/src/modules/certificates/completion-rules.ts    (new)
lmshub-be/src/modules/courses/courses.controller.ts
lmshub-be/src/modules/courses/courses.repository.ts
lmshub-be/src/modules/courses/courses.routes.ts
lmshub-be/src/modules/courses/courses.service.ts
lmshub-be/src/modules/courses/courses.validation.ts
lmshub-be/src/modules/documents/documents.controller.ts
lmshub-be/src/modules/documents/documents.repository.ts
lmshub-be/src/modules/documents/documents.routes.ts
lmshub-be/src/modules/documents/documents.service.ts
lmshub-be/src/modules/documents/documents.validation.ts
lmshub-be/src/modules/enrollments/enrollments.controller.ts
lmshub-be/src/modules/enrollments/enrollments.repository.ts
lmshub-be/src/modules/enrollments/enrollments.routes.ts
lmshub-be/src/modules/enrollments/enrollments.service.ts
lmshub-be/src/modules/progress/progress.repository.ts
lmshub-be/src/modules/progress/progress.service.ts
lmshub-be/src/modules/site-content/site-content.controller.ts
lmshub-be/src/modules/site-content/site-content.routes.ts
lmshub-be/src/modules/site-content/site-content.service.ts
lmshub-be/src/modules/users/users.controller.ts
lmshub-be/src/modules/users/users.repository.ts
lmshub-be/src/modules/users/users.routes.ts
lmshub-be/src/modules/users/users.service.ts
lmshub-be/src/modules/users/users.validation.ts
lmshub-be/src/scripts/set-admin-email.ts    (new)
lmshub-be/tests/unit/completion-rules.test.ts    (new)
lmshub-be/tests/unit/content-safety.test.ts    (new)
```

Frontend — 66 files (translation files are grouped per language):

```
lmshub-fe/src/components/ui/RichTextEditor.vue    (new)
lmshub-fe/src/config/nav.ts
lmshub-fe/src/layouts/PublicLayout.vue
lmshub-fe/src/lib/api.ts
lmshub-fe/src/modules/assessments/assessments.routes.ts
lmshub-fe/src/modules/assessments/useCourseOptions.ts    (new)
lmshub-fe/src/modules/assessments/views/AssessmentListView.vue
lmshub-fe/src/modules/assessments/views/QuestionBankView.vue    (new)
lmshub-fe/src/modules/assessments/views/QuizAttemptView.vue
lmshub-fe/src/modules/assessments/views/QuizBuilderView.vue
lmshub-fe/src/modules/catalog/views/StaticPageView.vue    (new)
lmshub-fe/src/modules/certificates/views/MyCertificatesView.vue
lmshub-fe/src/modules/courses/views/CourseEditorView.vue
lmshub-fe/src/modules/learn/views/CoursePlayerView.vue
lmshub-fe/src/modules/settings/components/BrandIdentityCard.vue
lmshub-fe/src/modules/users/views/ProfileView.vue
lmshub-fe/src/modules/website/views/PagesView.vue    (new)
lmshub-fe/src/modules/website/views/WebsiteView.vue
lmshub-fe/src/modules/website/website.routes.ts
lmshub-fe/src/router/index.ts
lmshub-fe/src/stores/siteContent.ts
lmshub-fe/src/style.css
lmshub-fe/src/i18n/messages/{ar,en,hi,id}/assessments.json
lmshub-fe/src/i18n/messages/{ar,en,hi,id}/certificates.json
lmshub-fe/src/i18n/messages/{ar,en,hi,id}/common.json
lmshub-fe/src/i18n/messages/{ar,en,hi,id}/courses.json
lmshub-fe/src/i18n/messages/{ar,en,hi,id}/errors.json
lmshub-fe/src/i18n/messages/{ar,en,hi,id}/learn.json
lmshub-fe/src/i18n/messages/{ar,en,hi,id}/nav.json
lmshub-fe/src/i18n/messages/{ar,en,hi,id}/pages.json    (new)
lmshub-fe/src/i18n/messages/{ar,en,hi,id}/settings.json
lmshub-fe/src/i18n/messages/{ar,en,hi,id}/users.json
lmshub-fe/src/i18n/messages/{ar,en,hi,id}/website.json
```

Also in the package: [LMSHub-Update-Guide.pdf](#) (this guide) and [SHA256SUMS.txt](#).